HDCGNet: A Hypergraph Dual-Branch CNN–GCN Network for Mars Hyperspectral Image Classification

Anhong Tian, *Member, IEEE*, Tao Chen, Sen Lei, *Member, IEEE*, Chengbiao Fu, *Member, IEEE*, Huaiping Jin, *Member, IEEE*, and Zhenwei Shi, *Senior Member, IEEE*

***Abstract*—The hyperspectral image (HSI) classification, as an important research direction in the field of remote sensing, has made significant progress in Earth observation. However, its application in Mars exploration missions is still in the exploratory stage. While convolutional neural networks (CNNs) are effective in extracting local features, their inherent local receptive field restricts the ability to capture long-range spatial dependencies. To overcome this limitation, we propose a dual-branch fusion architecture, termed HDCGNet, which seamlessly integrates CNNs and graph convolutional networks (GCNs), and further incorporates hypergraph learning into the GCN branch. This design enables the extraction of both local and global information from HSI and facilitates the effective modeling of high-order nonlinear relationships among multiple nodes. First, a cascade processing structure is employed in the CNN feature extraction branch to perform spectral denoising and transformation. By leveraging depthwise separable convolution (DSC) and cross multiscale convolution modules, the model effectively captures multiscale spatial–spectral joint features and enhances pixel-level feature representation with differentiated receptive fields. Second, to enrich the topological features of the HSI segmentation area, a hypergraph is added to the GCN feature modeling branch, while the ContraNorm normalization layer ensures a uniform distribution of the representation space and facilitates hyperpixel-level feature extraction. Finally, the adaptive cross-attention fusion module (ACAFM) is employed to fuse the features of the two branches, thereby ensuring complementarity between global and local information. Experimental results on three Mars HSI datasets demonstrate that HDCGNet achieves better performance than some existing methods. Our codes and data will be made publicly available at: https://github.com/Ctao0820/HDCGNet.git**

***Index Terms*—Convolutional neural network (CNN), deep learning, graph convolutional network (GCN), hypergraph learning, hyperspectral image (HSI) classification, Martian minerals.**

I. Introduction

SINCE the 1960s, continuous advances in space probe technology have enabled the acquisition of multisource data from orbiters, landers, and rovers, thereby laying the foundation for applying deep learning to Martian surface analysis [1], [2]. Hyperspectral imaging technology, characterized by hundreds of continuous spectral bands, enables accurate identification of mineral distributions on the Martian surface [3], [4], which is crucial for investigating the geological evolution of Mars. Recently, Martian mineralogy research has entered the stage of multidimensional information fusion. Hyperspectral classification has emerged as an essential technique for investigating the geological evolution of Mars, since it provides simultaneous access to information on mineral composition and spatial distribution. Traditional spectral matching methods based on distance metrics are computationally efficient but are highly susceptible to noise and mixed pixels. Consequently, researchers have developed methods such as probabilistic models [5], unsupervised learning, and deep neural networks [6], which have significantly improved the recognition accuracy. However, deep learning methods continue to encounter challenges, including the limited availability of annotated data on Mars [7] and the high spectral similarity of Martian minerals, which constrain the applicability of these models.

Hyperspectral data contains rich spectral and spatial information [8], [9], [10] and has been widely used in the fields such as agriculture [11], urban planning [12], environmental monitoring [13], marine [14], and deep space exploration [15]. Hyperspectral image (HSI) classification is the core technology in these applications, as it analyzes the unique spectral characteristics of each pixel to achieve accurate identification and classification of land cover types. In addition, since HSIs usually contain hundreds of thousands to millions of pixels, directly constructing a network with a single pixel as a graph

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Anhong Tian is with the Faculty of Information Engineering and Automation, Kunming University of Science and Technology, Kunming 650500, China, also with the Faculty of Land Resources Engineering, Kunming University of Science and Technology, Kunming 650093, China, and also with the Yunnan Key Laboratory of Intelligent Monitoring and Spatiotemporal Big Data Governance of Natural Resources, Kunming 650093, China (e-mail: tah@kust.edu.cn).

Tao Chen, Chengbiao Fu, and Huaiping Jin are with the Faculty of Information Engineering and Automation, Kunming University of Science and Technology, Kunming 650500, China (e-mail: taochen@stu.kust.edu.cn; fcb@kust.edu.cn; jinhuaiping@gmail.com).

Sen Lei is with the School of Information Science and Technology, Southwest Jiaotong University, Chengdu 611756, China (e-mail: senlei@swjtu.edu.cn).

Zhenwei Shi is with the Department of Aerospace Intelligent Science and Technology, School of Astronautics, and the State Key Laboratory of Virtual Reality Technology and Systems, Beihang University, Beijing 100191, China (e-mail: shizhenwei@buaa.edu.cn).

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node will face serious computational complexity problems. To solve this problem, the current mainstream methods use the simple linear iterative clustering (SLIC) hyperpixel segmentation technique [16] to preprocess HSI, which aggregates adjacent pixels with similar spectral features into hyperpixel units, thereby significantly reducing the size of the graph. Wan et al. [17] proposed a hyperpixel generation method based on spectral–spatial joint measurement, which combines spectral similarity with spatial proximity to achieve adaptive regional aggregation of HSI data, significantly improving the efficiency of subsequent graph convolution operations.

Convolutional neural network (CNN), as the most commonly used deep learning model [18], has proven its effectiveness in HSI classification [19], [20]. In the development of HSI classification, from the early 1DCNN [21] that focused on spectral features, to the 2DCNN [22] that introduced spatial information, and then to the 3DCNN [23] that jointly extracted spectral and spatial features, CNNs have played a crucial role. Researchers have improved the performance of CNN through a variety of innovative methods. Zhong et al. [24] proposed an end-to-end spectral–spatial residual network (SSRN) by incorporating a residual learning mechanism, which can effectively capture a wide range of spectral features and spatial contextual discriminant information in hyperspectral data. Roy et al. [25] proposed an adaptive spectral–spatial kernel improved residual network (A^2S^2K-ResNet) based on the attention mechanism, which innovatively integrated the attention mechanism with adaptive kernel learning and effectively improved the ability to extract discriminative spectral–spatial features. CNNs have certain key limitations in HSI classification: on the one hand, the local receptive field restricts global context modeling; on the other hand, short-range convolution hinders the extraction of highly discriminative spatial–spectral features.

In contrast, the graph convolutional network (GCN) generalizes the convolution operation to non-Euclidean space, enabling effective feature extraction from graph data with arbitrary topological structures [26] and thereby making it suitable for HSI tasks. GCN maps pixels to graph nodes and builds edges based on spectral similarity to achieve global spatial–spectral feature modeling. It can also use the graph structure to alleviate overfitting of small samples. Ding et al. [27] used a multiscale graph sampling and aggregation network architecture (GraphSAGE) based on GCN to extract multiscale features in layers and utilize a context-aware mechanism to describe the importance of spatially adjacent regions. Similarly, the attention mechanism has shown significant advantages in improving the performance of GCNs. The graph attention network (GAT) [28] achieves dynamic adaptive adjustment of the connection weights between nodes by introducing the self-attention mechanism. Han et al. [29] designed a semi-supervised classification method based on the multiscale spectral spatial GAT (MSSGAT), which constructs neighborhood graphs of multiple scales, where all samples are represented as nodes, and utilizes spectral spatial features with different receptive fields.

CNN is effective in extracting local pixel features, whereas GCN is capable of capturing global hyperpixel features. However, their direct fusion encounters several challenges. First, the data structure is incompatible, which leads to information loss during the conversion process between regular grid features and irregular graph structures. Second, there is a dimension mismatch problem, which increases the computational overhead during data conversion. Finally, the training is dynamically unbalanced, and the difference in subnetwork convergence speed affects the overall performance. To solve the above problems, Liu et al. [30] proposed a CNN-enhanced GCN (CEGCN), which achieved deep integration of CNN and GCN by designing a dual-path encoding–decoding architecture. Wang et al. [23] designed and proposed a new HGCN-MHF for HSI classification with limited samples, introduced a hypergraph mechanism, and achieved good small sample classification results. Based on the previous analysis, we proposed a hybrid CNN–GCN network (HCGN) for HSI classification on Mars with limited samples.

The network architecture of this study includes a two-branch feature extraction module: in the CNN branch, we first use the spectral domain denoising layer combined with 1-D convolution to enhance the spectral features, and combine it with deep separable convolution to reduce the amount of data. Then, multigranularity spatial feature extraction is achieved through the cross-scale multiconvolution module. In the GCN branch, this study proposes a hypergraph convolutional network architecture, the core of which is to deeply explore the spatial–spectral collaborative representation capabilities in hyperspectral data. The hypergraph is composed of three hyperedge construction strategies. In the construction of kNN hyperedges, SSIM is innovatively introduced as an evaluation index. Compared with the traditional Euclidean distance, the SSIM index can better maintain the structural characteristics of the spectral curve and has stronger robustness to illumination changes and noise interference. The ContraNorm normalization layer is used in the normalization layer to propagate hyperpixel-level features. Finally, the adaptive cross-attention fusion module (ACAFM) is referenced, and the channel attention cross module and the spatial attention fusion module are integrated to perform adaptive feature fusion on the dual-branch output results. In general, the main contributions of this article are summarized as follows.

1) To employ a depthwise separable module in the multiscale CNN branch for preserving channel-wise information and minimizing computational cost, and to realize pixel-level multigranularity feature capture through parallel processing of three convolutional kernels with different sizes.
2) To introduce a GCN branch based on spatial-domain hierarchical map convolution, which integrates hyperpixel composition and a neighborhood aggregation mechanism to effectively extract region-level contextual features, and to incorporate a ContraNorm normalization layer for maintaining a uniform distribution in the representation space.
3) To achieve adaptive processing of the two branches using ACAFM, and to facilitate cross-scale complementation between the branches through the feature alignment module.

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4) To apply the proposed model for the classification of Mars HSIs and to conduct mineral classification in selected Martian regions.

The rest of this article is organized as follows. Section II provides a brief overview of related work. Section III introduces the proposed HDCGNet in detail. Section IV presents the experiments and analyzes the classification results. Section V draws conclusion.

II. Related Work

A. CNN-Based HSI Classification Method

In the past decade, CNNs have made significant progress in the field of HSI classification due to their powerful feature representation capabilities. This deep learning architecture automatically learns discriminative spatial–spectral features through hierarchical convolution operations, changing the paradigm of traditional classification methods. Ma et al. [31] proposed a lightweight hybrid CNN (LiteHCNet) for HSI classification, minimizing model parameters and computational workload. Zheng et al. [32] proposed a lightweight end-to-end attention-enhanced feature fusion network (AeF2N) to better understand the inherent spectral complexity and spatial variability in HSI. Dong et al. [33] proposed an HSI classification method based on small sample data expansion and consistency regularization, using a designed CNN-based model to extract discriminative features from a limited number of labeled pixel-pair features (PPFs) and a large number of unlabeled PPFs. In addition, attention mechanisms have been widely introduced into CNNs. For example, many attention mechanisms, such as the hybrid attention mechanism convolutional block attention module (CBAM) [34] and the channel attention mechanism (ECA) [35] have been adopted and applied. Cui et al. [36] designed a dual-triple attention network (DTAN) based on capturing cross-dimensional interaction information. The ECA module was introduced into the DenseNet module to achieve partial cross-channel interaction. Although CNN-based HSI classification methods have demonstrated excellent performance in multiple tasks, their inherent architecture still makes it difficult to effectively model the long-range dependencies between spatial and spectral dimensions in hyperspectral data.

B. GCN-Based HSI Classification Method

In contrast to CNNs, which mainly focus on local feature extraction, GCNs and their derivative models leverage graph-structured data to capture long-range spatial–spectral dependencies that CNNs fail to represent, while effectively modeling the topological relationships among spatial nodes. Yang et al. [37] proposed a classification model, DMSGer, based on the GCN paradigm. They constructed a relatively small graph at the region level based on the hyperpixel segmentation algorithm and metric learning, and then applied the dynamic pixel-level feature update strategy to the region-level adjacency matrix to achieve the purpose of dynamically learning pixel representation. Wang et al. [38] proposed a dual-branch hybrid architecture DBMGNet that combines Mamba and GCN, designed the band selection enhanced bidirectional Mamba (BSEBM), and applied reparameterized Chebyshev graph convolution (RCGC) in GCN to simulate the similarity dependency in non-Euclidean space, while designing the adjacency matrix based on the intrinsic characteristics of HSI. Chen et al. [39] proposed an automatic graph learning convolutional network (Auto-GCN), which unified graph learning and HSI classification in a "network" manner to model the interactions of high-order tensors. GCN can perform convolution operations on graphs of arbitrary topological structures, convert HSI data into graph structure representations, and model the correlation between different land covers and HSI spatial topologies.

C. Hypergraph Learning

GCN is mainly divided into two processing paradigms according to the differences in task objectives: the first is graph-level classification, which focuses on the global representation of the entire graph structure; the second is node-level classification, which mainly focuses on the feature learning of individual nodes in the graph. In the HSI classification task, by mapping pixels or hyperpixels into graph nodes and building edge connections based on spectral–spatial similarity, it can be converted into a node classification problem under the GCN framework [40]. Currently, most GCN methods regard each pixel as a node and the spectral information of each pixel as the node feature, thereby constructing graph data from HSI using a specific distance measurement method. Current GCN-based methods still have some defects. Existing methods rely heavily on the construction of the adjacency matrix, which represents the topological structure of graph data. However, the adjacency matrix is basically based on prior knowledge and cannot be updated through the learning process. In addition, using a simple graph structure for modeling, where edges can only represent pairwise relationships, fails to discover the underlying complex structure in the data [41].

Based on the above problems, hypergraphs can be introduced into GCN. A hypergraph is a generalized graph structure that contains a set of vertices and hyperedges. A hyperedge is set between multiple vertices to represent a higher order relationship. The number of vertices is usually greater than 2. Hypergraph learning is generalized from graph learning, and its purpose is to analyze data by learning the information transfer mode between hypergraph structures. Feng et al. [42] first proposed HGNN and introduced the spectral convolution operator on the static hypergraph. Hypergraph learning can establish high-order correlation models between data, so hypergraph learning can well capture the nonlinear high-order data in HSIs. Currently, there are few studies on the use of HGNN for HSI classification. For example, Duan et al. [43] proposed a structure-preserving hypergraph convolutional network SPHGCN, which integrates local regular convolution and irregular hypergraph convolution to learn the structured semantic features of HSI. Xu et al. [44] proposed a dynamic hypergraph convolution and recursive gated convolution fusion network for HSI classification to learn the hidden and important relationships represented in HSI data.

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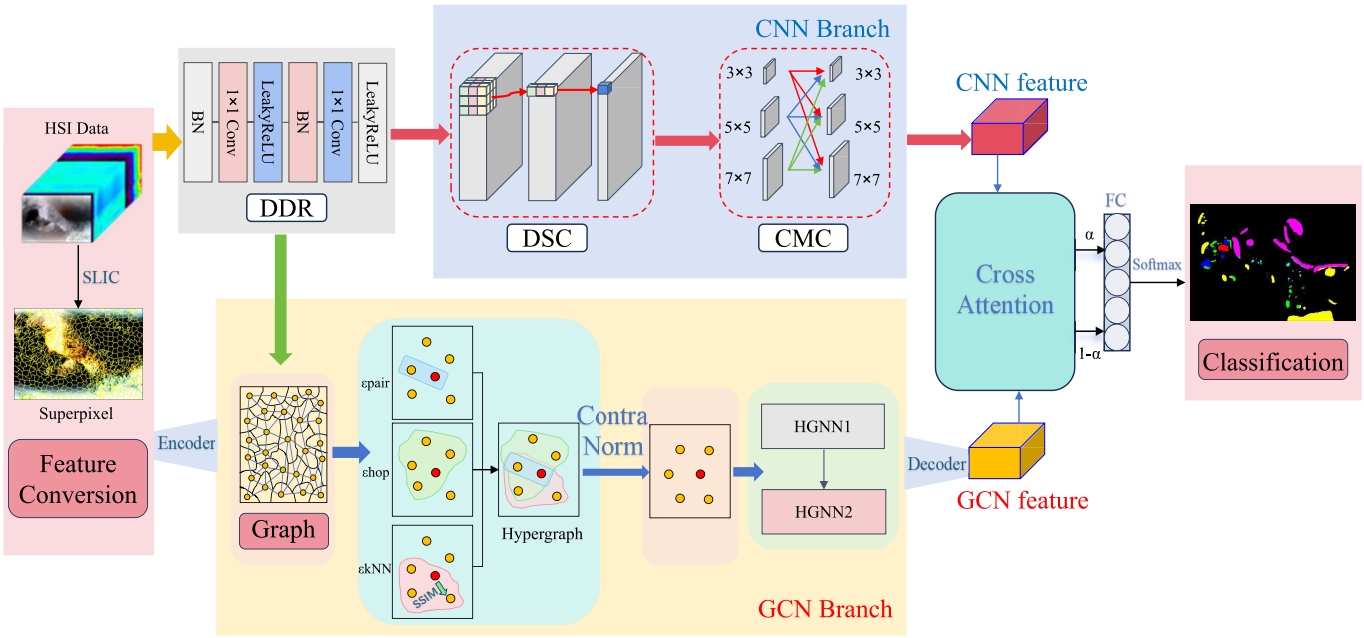


Fig. 1. Framework of the proposed HDCGNet.

D. Dual-Branch CNN–GCN

The current HSI classification methods based on deep learning still have obvious limitations in feature extraction. Specifically, whether it is CNN or GCN architecture, when only a single-layer convolutional structure is used, its information-capturing ability will be limited by the inherent receptive field. For CNN-based methods, it usually constructs the input patch by intercepting fixed-size neighborhood pixels centered on the sampling point. This processing method has two key defects: first, the fixed-size local receptive field cannot adapt to the changing requirements of multiscale features in HSI data; second, this rigid sampling strategy will ignore the optimal feature scales unique to different types of objects, resulting in poor classification performance of some objects with special scale characteristics. Similarly, GCN-based methods also have significant constraints: the network can only learn on a predefined single graph structure, and each hyperpixel node can only aggregate the information of its direct neighboring nodes. This local information transmission mechanism severely limits the network's ability to capture long-range spatial dependencies. These limitations fundamentally restrict the classification performance of existing methods in complex scenes, and there is an urgent need to develop a new network architecture that can adapt to multiscale features and effectively integrate global context information.

These shortcomings prompted researchers to adopt a dual-branch CNN–GCN network architecture design. This architecture effectively integrates the advantages of the two networks through a carefully designed feature complementation mechanism: the CNN branch is good at extracting local spectral–spatial features, and its hierarchical convolutional structure can automatically learn multilevel feature representations from low-level textures to high-level semantics; while the GCN branch focuses on modeling global topological relationships and captures long-range spatial dependencies between pixels through graph convolution operations. In particular, the two branches achieve deep complementarity through an innovative feature interaction mechanism: on the one hand, the local features extracted by CNN provide a reliable basis for GCN to construct neighborhood relations; on the other hand, the global context information learned by GCN supplements CNN with semantic associations beyond the local receptive field. This synergy enables the network to adaptively process features of objects at different scales, enhancing global consistency while maintaining local detail accuracy, thereby significantly improving classification performance in complex scenarios. Compared with a single network, this dual-branch architecture can more fully mine the multiscale spatial–spectral information contained in HSI data. Shi et al. [45] proposed an HCGN for HSI classification that combines the advantages of CNN and GCN frameworks and can provide excellent classification performance with limited training samples. Zhou et al. [46] designed an attention multihop graph and multiscale convolutional fusion network (AMGCFN) that is conducive to the learning of multilayer contextual semantic features. Yu et al. [47] proposed a band selection method to collect HSI classification discriminant band subsets, which developed GCN and CNN to refine the band correlations sequentially. The bands selected according to the obtained correlations were fed to the proposed self-supervised auxiliary spectral reconstruction network for spectral reconstruction to achieve band selection.

III. Proposed Method

In this section, we introduce the proposed HDCGNet model in detail. The proposed HDCGNet is a hybrid neural network

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that performs the extraction and fusion of hyperpixel and pixel features to characterize the spectral and spatial properties of HSI. The constructed network mainly includes two branches. One is the CNN branch, which is used to extract pixel-level features, reduce computational complexity, reduce information redundancy, optimize the network structure, and prevent overfitting. The other is the GCN branch. The advantage of this branch design is that it makes full use of the inherent spatial topological structure and spectral characteristics within the segmented area, constructs a multilevel feature expression system through a hypergraph modeling method, extracts the spatial adjacency matrix within the segmentation unit, integrates the spectral similarity measure to construct hyperedge connections, and establishes a hypergraph topological structure with physical significance for extracting hyperpixel-level features. Finally, the features of the two branches are introduced in channel attention and spatial attention for adaptive fusion. In the following, we will describe each module in detail. The complete architecture of our model is shown in Fig. 1. The HSI data is first denoised and dimensionally reduced by the DDR module, and then pixel-level features are obtained through the DSC module and CMC multiscale parallel in the CNN branch. Then, from the dimensionally reduced data and the input GCN branch after SLIC, after multistrategy hypergraph processing, feature stability is achieved through the ContraNorm normalization layer, and after two layers of HGNN, hyperpixel-level features are obtained. Ultimately, the two adaptively cross-fuse to obtain the final output result.

A. Hypergraph Convolution Composed of Multiple Hyperedges

In the HSI classification task, traditional hypergraph learning methods have certain limitations. The hyperedge generation strategy has certain defects, lacks the ability of other methods to generate hyperedges, relies heavily on the kNN algorithm to construct hyperedges, pays too much attention to spectral features and ignores the spatial topological relationship of hyperpixel regions, and fails to effectively utilize the spatial structure information provided by image segmentation. Second, there is a problem of feature distortion. In addition to the noise interference inherent in HSI data, the mixed pixel effect in the hyperpixel segmentation area will cause deviations in spectral feature representation. Finally, the hyperedge constructed solely based on spectral similarity may misrepresent the actual spatial correlation between objects, affecting the accuracy of subsequent feature propagation.

To address the limitations of traditional hypergraph construction, we propose a multistrategy hypergraph fusion module, whose core architecture consists of three parallel hyperedge generation strategies: pairwise hyperedges ($\varepsilon_{\text{pair}}$), k-hop hyperedges ($\varepsilon_{\text{hop}}$), and kNN hyperedges ($\varepsilon_{\text{kNN}}$).

The construction of the graph structure in this study begins with the hyperpixel generation stage. The SLIC algorithm is used to over-segment the original HSI data $X \in \mathbb{R}^{H\times W\times C}$ to generate Z spatially compact and spectrally uniform hyperpixel regions, making each hyperpixel a node in the graph structure. Second, define the mapping matrix ($H \times W$) to represent the pixel-hyperpixel correlation, where HW represents the total number of pixels in the input image ($H \times W$), Z represents the total number of hyperpixels generated, and the matrix element $Q_{i,j}$ is expressed as

$$Q_{i,j} = \begin{cases} 1, & \text{if } X_i \in S_j \\ 0, & \text{otherwise} \end{cases} \quad X = Flatten(X) \tag{1}$$

where $Q_{i,j}$ reflects the relationship between the ith pixel element $\widehat{X}_i$ and the jth hyperpixel element S_j. $Flatten(\cdot)$ means flattening X in space. $\widehat{X}_i$ represents the ith pixel in $\widehat{X}$, and S_j represents the jth hyperpixel. We can then use the following encoder and decoder to achieve the conversion between nodes and hyperpixels, which are defined as follows:

$$V = Encoder(X, Q) = \hat{Q}^T\, Flatten(X) \tag{2}$$

$$\hat{X} = Decoder(X, Q) = Reshape(QV) \tag{3}$$

where $\hat{Q}^T$ represents the transpose of $\hat{Q}$ after column normalization. V represents the feature matrix of all hyperpixel nodes. $Reshape(\cdot)$ represents the spatial dimension of the restored flat data. $Encoder(\cdot)$ represents the conversion of pixel graph into graph nodes, and $Decoder(\cdot)$ represents the mapping of hyperpixel nodes to pixels.

Based on the undirected graph structure $G = (V, \varepsilon)$ obtained after hyperpixel segmentation, the hypergraph is constructed. Undirected edges are introduced to connect adjacent hyperpixels in the spatial distribution, and three hyperedge construction strategies are used to build the hypergraph. The first strategy is to use paired hyperedges ($\varepsilon_{\text{pair}}$) to directly transform all undirected edges in the base graph into binary hyperedges, that is, each hyperedge has only two nodes and is built based on the direct connection edges between nodes, retaining the topological relationship of the original graph structure, preserving local spatial neighborhood information, and constructing low-order neighbor edge relationships to provide a basis for high-order relationships. The second strategy is the k-hop hyperedge (ε_{hop}), which captures multiorder neighborhood relationships. It takes the central node v_i as the starting point and connects all nodes in its k-hop neighborhood through k-hop path extension, enhancing the long-range dependency modeling capability and capturing multiscale spatial context information. The third strategy is the k-nearest neighbor hyperedge (ε_{kNN}), which is constructed based on the structural similarity index (SSIM) as a similarity metric. It constructs a hyperedge by selecting the nodes with the greatest structural similarity in the constrained spatial neighborhood to obtain local spectral information. It only requires hyperpixel node information and avoids noise interference caused by a global search. We linearly adjust SSIM to $[-1, 1]$ to ensure compatibility with spatial weights. SSIM is defined as

$$SSIM(x, y) = \left[l(x, y)\right]^{\alpha} \cdot \left[c(x, y)\right]^{\beta} \cdot \left[s(x, y)\right]^{\gamma} \tag{4}$$

where $l(x,y)$ denotes luminance comparison, $c(x,y)$ denotes contrast comparison, $s(x,y)$ denotes structural comparison, and α, β, and γ denote structural parameters. We convert the features of each hyperpixel pair into image format, calculate their SSIM value, and finally return the obtained value.

After generating multiple sets of hyperedges, this module introduces an adaptive weight function to achieve feature

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fusion, and the fusion criterion considers both spectral feature similarity as well as spatial neighborhood distance. The spectral similarity of the nodes v_i and v_j is as follows:

$$\phi_{\text{spectral}}\left(V_i, V_j\right) = \exp\left(-\frac{\left(F_i - F_j\right)^2}{2\sigma_1^2}\right) \tag{5}$$

where ℓ_2 is the L2 paradigm, F_i and F_j denote the spectral features of nodes V_i and V_j, respectively, and σ_1 is the adjustable parameter in the feature similarity measure function. Similarly, the spatial similarity is defined as follows:

$$\phi_{\text{spatial}}\left(V_i, V_j\right) = \exp\left(-\frac{\left(P_i - P_j\right)^2}{2\sigma_2^2}\right) \tag{6}$$

where P_i and P_j denote the central position of the hyperpixel nodes V_i and V_j, respectively. σ_2 is an adjustable parameter in the spatial distance measurement function. A hyperedge can connect multiple nodes. The spectral measurement weights and spatial measurement weights of the hyperedge ε_k are designed as follows:

$$W_{\text{spectral}}\left(\varepsilon_k\right) = \frac{\sum \phi_{\text{spectral}}\left(V_i, V_j\right)}{C\left(|\varepsilon_k|, 2\right)} \tag{7}$$

$$W_{\text{spatial}}\left(\varepsilon_k\right) = \frac{\sum \phi_{\text{spatial}}\left(V_i, V_j\right)}{C\left(|\varepsilon_k|, 2\right)} \tag{8}$$

where $V_i, V_j \in \varepsilon_k$ and $i \neq j$, $|\varepsilon_k|$ denotes the number of nodes in the hyperedge, and $C(|\varepsilon_k|, 2)$ denotes the number of combinations of two different elements selected from the hyperedge. Finally, the weights of the hyperedge ε_k are calculated as follows:

$$W_{\text{spa-spe}}\left(\varepsilon_k\right) = \min\left(W_{\text{spectral}}\left(\varepsilon_k\right), W_{\text{spatial}}\left(\varepsilon_k\right)\right) \tag{9}$$

where the minimization criterion $min(\cdot)$ determines the final weight of the hyperedge, which implements a double constraint mechanism, and the weight converges to 1 only when both spatial proximity and spectral similarity are satisfied, and this design has good robustness against outliers of a single mode. The measure of similarity between $[0, 1]$ is ensured by the RBF kernel function, which helps to control the weight of the hyperedge.

Compared with traditional GCN, the hypergraph CNN (HGNNConv) has significant advantages in graph data processing, especially in dealing with higher order associations and unpaired relations. Traditional GCNs can only handle binary edges, i.e., node pairs. This binary edge is only applicable to second-order neighborhood aggregation, but HGNNConv supports hyperedges, i.e., the ability to connect 2 and more nodes, which enables higher order interaction modeling. One of the hypergraph Laplace operators is

$$L = I - D_v^{-\frac{1}{2}} H W D_e^{-1} H^T D_v^{-\frac{1}{2}} \tag{10}$$

where D_v is the node degree matrix, D_e is the hyperedge degree matrix, and W is the hyperedge weight matrix, which supports spectral domain analysis of higher order relations. HGNNConv represents a spatial convolution method applied to hypergraphs, which is used in this study.

Given a hypergraph $G = (V, \varepsilon, W)$ that aggregates features from neighboring vertices, the central vertex feature

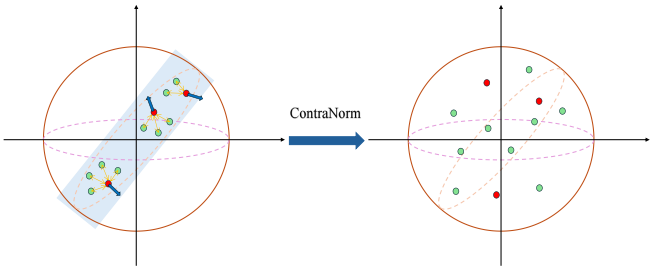


Fig. 2. Illustration of how ContraNorm solves dimensionality collapse.

`HGNNConv` is updated in a single phase to achieve feature propagation through two phases: vertex aggregation hyperedge and hyperedge update of new vertices, where $|V|$ is the number of vertices, $|\varepsilon|$ is the number of hyperedges, and the matrix $H \in \mathbb{R}^{|V| \times |\varepsilon|}$ represents the neighbor relationship in the hypergraph, which is defined as follows:

$$H(v, e) = \begin{cases} 1, & \text{if } v \in e \\ 0, & \text{if } v \notin e \end{cases}$$

where $v \in e$ represents the vertex v connected by the hyperedge e, and the vertex neighbor set $N_v(e)$ represents all nodes connected by the hyperedge e. In the first stage, the hyperedge feature of the lth layer can be expressed as

$$Y^l = W D_e^{-1} H^T V^l \tag{11}$$

where V^l denotes the hyperpixel vertex feature as input to the lth convolutional layer and W denotes the weight matrix of the hyperedge $W_{\text{spe-spa}}$. Similarly, in the second stage, the updated vertex feature V^{l+1} of the lth layer as the output can be expressed as

$$V^{l+1} = \sigma\left(D_v^{-1} H Y^l \Theta^l\right). \tag{12}$$

In this way, the matrix format of HGNNConv can be combined into a single formula

$$V^{l+1} = \sigma\left(D_v^{-1} H W D_e^{-1} H^T V^l \Theta^l\right). \tag{13}$$

The frequency-domain-based HGNNConv is defined as follows:

$$V^{l+1} = \sigma\left(D_v^{-\frac{1}{2}} H W D_e^{-1} H^T D_v^{-\frac{1}{2}} V^l \Theta^l\right). \tag{14}$$

This study consists of two HGNNConv layers. Each layer adopts the multistrategy hypergraph fusion mechanism described in the formula. The feature dimension is 128. When a hyperedge only connects two nodes ($\varepsilon_{\text{pair}}$), HGNNConv can be regarded as a special case of GCN.

B. ContraNorm Normalization Layer

To address the degradation of feature dimensionality in deep networks and maintain the uniform distribution of the representation space, and handle related issues such as over-smoothing that occur in HGNNConv, we adopt a nearly zero-overhead normalization layer design that implicitly contrasts the learning objectives to maintain the uniform distribution of the representation space and alleviate the key issue of feature dimensionality collapse. Dimensionality collapse

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refers to the fact that node features tend to be similar in deep networks and lose their discriminability, which is manifested as a decrease in the rank of the feature matrix. Deep stacking of layers will lead to indistinguishable representations and significant performance degradation, resulting in the occurrence of over-smoothing problems. By deeply analyzing the common characteristics of over-smoothing and representation collapse, a theoretical connection between the two is established. In addition, ContraNorm not only enhanced the feature discrimination ability of hypergraph nodes in this task, but also stabilized the high-order feature propagation process. An illustration of how ContraNorm solves dimensionality collapse is shown in Fig. 2.

The implementation of traditional uniformity loss relies on indirect optimization of model parameters, while the advantage of ContraNorm is that it transforms it into direct optimization of the representation itself. To achieve this goal, we first reformulate the uniformity loss at the training objective level as a representation constraint at the architecture level. Each node $h_i \in \mathbb{R}^d$ in a fully connected graph structure $G = (V, \varepsilon)$ corresponds to a representation vector of a sample. Inspired by the uniformity loss L_{uniform} in contrastive learning, it is generalized to a global loss function on the graph node representation

$$\hat{L}_{\text{uniform}} = \sum_{i \in V} \log \left(\sum_{j \in V} e^{\frac{h_i^T h_j}{\tau}} \right). \tag{15}$$

Taking the derivative of L_{uniform} for the node representation

$$\frac{\partial \hat{L}_{\text{uniform}}}{\partial h_i} = \sum_j \frac{\exp\left(\frac{h_i^T h_j}{\tau}\right)}{\sum_k \exp\left(\frac{h_i^T h_k}{\tau}\right)} \frac{h_j}{\tau} + \sum_j \frac{\exp\left(\frac{h_j^T h_i}{\tau}\right)}{\sum_k \exp\left(\frac{h_j^T h_k}{\tau}\right)} \frac{h_j}{\tau}. \tag{16}$$

Let the feature matrix be H, and $H > i$ in the ith row. We can rewrite the above formula into matrix form

$$\frac{\partial \hat{L}_{\text{uniform}}}{\partial h_i} = \left(D^{-1}A + AD^{-1}\right) H/\tau, \tag{17}$$

where $A = \exp(HH^T/\tau)$ and D denotes the diagonal degree matrix of A, with $D_{ii} = \sum_j A_{ij}$.

In order to maintain the stability of the representation, the choice is to use layer normalization to maintain the feature scale. The updated form of LayerNorm is

$$\text{LayerNorm}(h_i) = \gamma \cdot \left(\frac{h_i - \text{mean}(h_i)}{\sqrt{\text{Var}(h_i) + \varepsilon}} \right) + \beta \tag{18}$$

where γ and β are learnable parameters, $\varepsilon = 10^{-5}$. The learnable parameters γ and β can rescale the representation h_i to help alleviate the problem. Appending LayerNorm to H with a single-step gradient descent update yields

$$H_t = \text{LayerNorm}\left(H_b - \frac{s}{\tau} \times \text{softmax}\left(H_b H_b^T\right) H_b\right) \tag{19}$$

where H_b and H_t represent the representation before and after the update, respectively, and s is the step size of the gradient descent. Applying layer normalization to the representation matrix H means applying layer normalization to all its components $h_1, \ldots, h_n$, which allows ContraNorm to reduce the uniformity loss of the representation, thereby helping to alleviate dimensionality collapse.

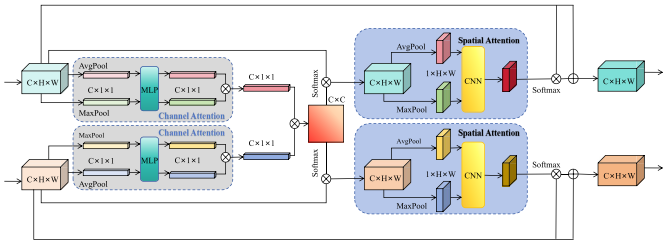


Fig. 3. CAFM structure diagram.

C. Adaptive Cross-Attention Fusion Module

In order to balance and promote the deep feature fusion of the two subnetworks, we use the ACAFM to process the two subnetworks. It uses cross-channel and spatial attention mechanisms to highlight salient features and enhance feature interactions between subnetworks, and maps the fused features to the final classification output through a fully connected layer. The model uses the focal loss function to ensure balanced training of the two subnetworks. The cross-attention fusion module (CAFM) structure is shown in Fig. 3. ACAFM consists of two parts: the channel attention cross module and the spatial attention fusion module. The features extracted by the CNN branch and the GCN branch are input separately, where global max pooling and average pooling are performed, and then they are placed in two layers of neural networks. The two obtained features are added together, passed through an activation function, to obtain the channel weights. Subsequently, the two fused features after spatial attention fusion are multiplied by the input features to obtain the fused features. Finally, an adaptive mechanism is added, and the fused features of the two parts are output through the fully connected layer.

CAFM consists of two parts: a channel attention cross module and a spatial attention fusion module. For the features $F^c \in \mathbb{R}^{C \times H \times W}$ from the two branches, they are first compressed along the spatial dimension by global maximum pooling (MaxPool) and global average pooling (AvgPool) to obtain two channel description vectors with the shape of $C \times 1 \times 1$, and extract pixel-level local features. Subsequently, these two description vectors are fed into a shared two-layer neural network (MLP), where the features of the two subnetworks are globally averaged and pooled, respectively, to generate channel description vectors. The channel weight is calculated by a shared two-layer MLP with ReLU activation, and the formula is

$$M^c = \sigma\left(\text{MLP}\left(\text{AvgPool}(F)\right) + \text{MLP}\left(\text{MaxPool}(F)\right)\right) \tag{20}$$

where $\sigma(\cdot)$ denotes the Sigmoid function used to normalize the weights to the range $[0, 1]$.

The channel weights M_T^c and M_H^c of the features T_{out} and H_{out} in the two subnets are obtained by the above method, and

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the characterization after the channel attention cross module is as follows:

$$T^c = \text{Softmax}\left(M_{cross}\right) T_{out} \tag{21}$$

$$H^c = \text{Softmax}\left(M_{cross}^T\right) H_{out} \tag{22}$$

$$M_{cross} = M_T^c \left(M_H^c\right)^T \tag{23}$$

where M_{cross} is the cross matrix of size $C \times C$ obtained by multiplying T_{out} and H_{out}. This matrix quantifies the correlation between the channels of the two subnetworks, with larger values indicating greater channel feature complementarity.

After Softmax normalization of the crossover matrix, for the fused features $F^s \in \mathbb{R}^{C\times H\times W}$ output from the channel attention module, we use a two-path pooling strategy to obtain the spatial attention cues. Specifically, we perform global MaxPool and average pooling along the channel dimension to obtain two spatial description maps of size $1 \times H \times W$, which are then concatenated to obtain enhanced spatial characterization. Then, the weight coefficients M^s are obtained by applying a shared convolutional layer with a kernel size of 3×3, as follows:

$$M^s = f^{3\times 3}\left(\text{AvgPool}\left(F\right) \oplus \text{MaxPool}\left(F\right)\right) \tag{24}$$

where $f^{3\times 3}$ denotes a convolutional layer with a kernel of 3×3.

Similarly, we can obtain the spatial weight coefficients M_T^s and M_H^s for the features T_{out} and H_{out}. By multiplying M_T^s and M_H^s with the corresponding input features, we obtain the fused features. In addition, we add residual connections to the fusion module to retain the original features while reinforcing the key regions. The specific formulation is as follows:

$$T^s = \text{Softmax}\left(M_T\right) T_{\text{out}} + T_{\text{out}} \tag{25}$$

$$H^s = \text{Softmax}\left(M_H\right) H_{\text{out}} + H_{\text{out}}. \tag{26}$$

We introduce adaptive adjustment in the output section. The fusion features T^s and H^s of the two parts are adaptively combined through a fully connected layer, and the final output Y is expressed as

$$Y = \text{Softmax}\left(W\left(\alpha T^s \oplus (1-\alpha) H^s\right) + B\right) \tag{27}$$

where W and B represent the weights and bias of the fully connected output layer, respectively, and the parameter α is constrained within the range $[0, 1]$. By default, $\alpha = 0.5$. It combines the outputs of CNN and GCN, fuses the weights α to balance the features of CNN and GCN, and activates them through Softmax$(\cdot)$.

Unlike the natural image task, the ACAFM in this study achieves adaptive alignment and complementarity of global topological features and local spectral–spatial features through dynamic weight learning.

In order to alleviate the category imbalance problem, the original cross-entropy loss is replaced by focal loss, which effectively addresses the common long-tailed distribution problem in HSI classification, improves the classification accuracy for small categories, and alleviates the category imbalance issue. The principle of **focal loss** is as follows:

$$L_{\text{Focal}} = -\sum_{i=1}^{N}\sum_{j=1}^{C} \alpha_j \left(1-p_t\right)^{\gamma} \log p_t \tag{28}$$

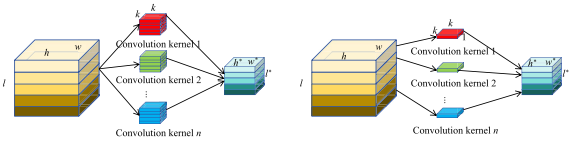


Fig. 4. (a) Standard 2-D convolution and (b) depthwise convolution.

where α_j denotes the category weight, γ is the focusing parameter, and p_t represents the predicted probability that the ith sample p_i belongs to the jth category.

D. Depthwise Separable Convolution (DSC)

CNN has gradually become the mainstream method in HSI classification tasks due to its excellent local feature extraction ability. However, HSIs have the characteristics of a large number of bands and rich information, which makes the traditional neural network model based on standard convolution often face the problem of too many parameters. In order to build a lighter network architecture, we use DSC to realize local feature extraction of HSIs. The structure consists of two parts: depthwise convolution and pointwise convolution. We set up two depth-separable convolutional blocks, each with 128 channels. Given an input feature map of size $h \times w \times l$, where h and w represent the spatial dimensions, and l is the number of spectral channels. The point convolution is then performed through n $1 \times 1 \times l$ convolution kernels to effectively model the nonlinear correlation between different spectral channels. Fig. 4 shows a schematic of conventional 2-D convolution and depthwise convolution.

Standard 2-D convolution uses n convolution kernels with a size of $k \times k$ $(k > 1)$ and a channel number of l to perform spatial–spectral joint feature extraction on the input feature map. The final calculation amount is

$$C_{std} = h \cdot w \cdot k^2 \cdot n \cdot l \tag{29}$$

where l is a loss function used to assess the differences between the predicted SLIC data and the ground truth. Depthwise convolution is a part of DSC, which uses l convolution kernels of size $k \times k$ and single channel to perform spatial convolution on each spectral channel of the input feature independently and then concatenates the results to output a feature map of $h \times w \times l$. The computational cost is

$$C_{dw} = h \cdot w \cdot k^2 \cdot l. \tag{30}$$

Pointwise convolution uses n $1 \times 1 \times l$ convolution kernels to fuse the features of different channels. The amount of calculation is

$$C_{pw} = h \cdot w \cdot n \cdot l. \tag{31}$$

Therefore, the total computational effort of DSC is

$$C_{dsc} = C_{dw} + C_{pw} = h \cdot w \cdot k^2 \cdot l + h \cdot w \cdot n \cdot l. \tag{32}$$

Compared with conventional 2-D convolution, the computational complexity of depth-separable convolution is

$$\frac{C_{dsc}}{C_{std}} = \frac{1}{n} + \frac{1}{k^2}. \tag{33}$$

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From the above formula, it can be concluded that DSC reduces the amount of computation compared to conventional 2-D convolution and can improve the running time of the model.

E. Other Modules

HSI data, due to its rich spectral information, has a dual nature: compared to multispectral imagery, its dense spectral features can significantly improve object classification accuracy; however, this information is often mixed with noise interference, which not only can affect classification accuracy but also imposes a huge computational burden. To this end, we adopt a dual 1×1 convolutional layer architecture. Through BatchNorm and LeakyReLU activation, it simultaneously achieves data denoising and feature dimensionality reduction. This approach effectively reduces the model's parameter count while significantly enhancing the system's robustness and operational stability. This module is called denoising and dimensionality reduction (DDR).

To enhance the feature extraction capability of the CNN-based network, we adopt a composite multiscale convolution (CMC) module, which includes three parallel branches with kernel sizes of 3×3, 5×5, and 7×7. This multiscale design is dedicated to generating rich pixel-level features. The CMC comprises two independent multiscale convolutional layers, each containing three convolutional blocks (ConvBlocks) with distinct receptive fields. After each multiscale convolutional layer, the features extracted from different kernel sizes are concatenated to facilitate cross-scale feature fusion.

Within each $k \times k$ ConvBlock, a batch normalization (BN) layer is first applied to normalize the data distribution, thereby accelerating training and mitigating overfitting. To reduce the spectral dimensionality of the high-dimensional features, a 1×1 convolutional layer is introduced after BN for feature transformation and parameter reduction. A LeakyReLU activation function is then applied to introduce nonlinearity. To further alleviate information redundancy, ease optimization, and suppress overfitting, an average pooling layer is incorporated. This pooling operation uses a 3×3 window with a stride of 1 and padding of 1 to preserve the spatial resolution of the feature maps. Finally, a 2-D CNN with a $k \times k$ kernel is employed to extract local spatial features

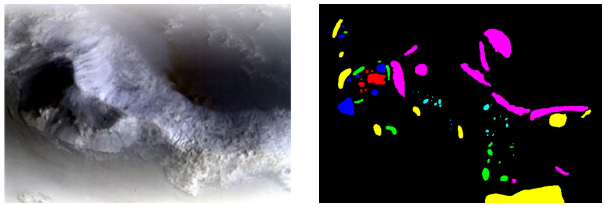
$$X_{i,j}^{l} = \sigma\left(\sum_{h=0}^{k-1}\sum_{w=0}^{k-1} W_{h,w}^{l} X_{i+h,j+w}^{l-1} + B_{i,j}^{l}\right) \tag{34}$$

where $X_{i,j}^{l}$ represents the output feature of the lth layer at the spatial position (i, j), $W_{h,w}^{l}$ represents the weight of the kernel, $X_{i+h,j+w}^{l-1}$ represents the input feature of the lth layer after the spatial position $(i+h, j+w)$ average pooling operation, $B_{i,j}^{l}$ is the bias, and the final activation function $\sigma(\cdot)$ is the $LeakyReLU(\cdot)$ of ConvBlock.

The workflow of HDCGNet is shown in Algorithm 1.

Algorithm 1 HDCGNet

Input: Hyperspectral image $\mathbf{X}$; Number of epochs E.
Output: Classification results $\mathbf{C}$.
Initialization: $e \leftarrow 0$.
Denoising and dimensionality reduction, and construction of a graph based on hyperpixels.
for $e = 1$ to E do
  Perform depthwise separable convolution and composite multi-scale convolution on $\mathbf{X}_1$, obtain local features $\mathbf{L}$ using Eq. (34).
  Generate hyperedge weights using Eqs. (7)–(9).
  Construct the hypergraph by applying Eq. (12) to $\mathbf{X}_2$ after SLIC processing.
  Alleviate over-smoothing using Eq. (19) and obtain global features $\mathbf{G}$.
  Fuse $\mathbf{L}$ and $\mathbf{G}$ using Eqs. (25) and (26).
  Adaptively output feature $\mathbf{T}$ through the fully connected layer using Eq. (27).
  Compute the loss using Eq. (28).
  Update network parameters using the Adam optimizer.
end
Obtain the classification results $\mathbf{C}$ from feature $\mathbf{T}$.



(a) (b)

Fig. 5. HC dataset. (a) False-color image. (b) Ground-truth map.

IV. Experimental Results and Analysis

A. Datasets, Parameter Settings, and Evaluation Metrics

To evaluate the effectiveness of our proposed method, we employed three Mars hyperspectral datasets[1] to validate the proposed method, which include scenes from Holden Crater (HC), Nili Fossae (NF), and Utopia Planitia (UP). The data were acquired by the Compact Reconnaissance Imaging Spectrometer for Mars (CRISM) on the Mars Reconnaissance Orbiter (MRO). CRISM data have a high spatial resolution of 18 m/pixel and include 544 spectral bands ranging from 0.36 to 3.93 μm, which are particularly suitable for mineral detection on the Martian surface.

1) Holden Crater: HC is known for its fluvial geological evolution, although the details of this evolution, such as water level changes and timing, remain unclear. The HC dataset consists of 418×595 pixels, and 440 bands are retained after filtering out water absorption and noise spectra. The false-color image and ground-truth image of the HC dataset are shown in Fig. 5, which contains six mineral categories: analcime, plagioclase, prehnite, high-ca pyroxene, serpentine, and margarite. The category name, detailed sample number, and labeled color are given in Table I.

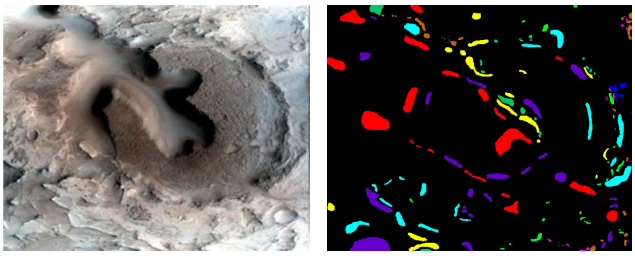
2) Nili Fossae: NF is unique in that it contains additional mineral phases such as carbonates and serpentine, which indicates the possibility of hydrothermal activity—an important factor for the possible occurrence of biological activity on Mars. The NF dataset has a size of 478×593 pixels and includes 425 bands after discarding several noisy

[1] Available: https://doi.org/10.57760/sciencedb.19732

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TABLE I
CATEGORY NAMES AND SAMPLE NUMBERS IN THE HC DATASET

Class ID	Color	Name	Train	Val	Test
1		Analcime	10	1	929
2		Plagioclase	10	1	1461
3		Prehnite	10	1	1549
4		High-Ca Pyroxene	10	1	6952
5		Serpentine	10	1	8686
6		Margarite	10	1	447
Total			60	6	20024



(a) (b)

Fig. 6. NF dataset. (a) False-color image. (b) Ground-truth map.

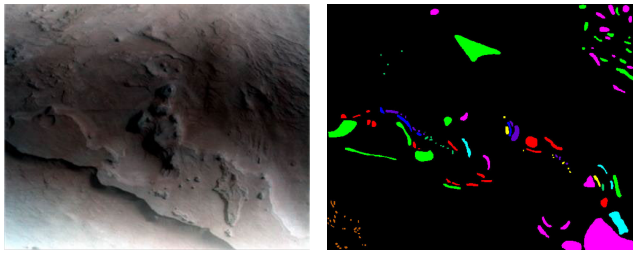
TABLE II
CATEGORY NAMES AND SAMPLE NUMBERS IN THE NF DATASET

Class ID	Color	Name	Train	Val	Test
1		Fe-Olivine	10	1	8552
2		Epidote	10	1	985
3		Chlorite	10	1	375
4		Bassanite	10	1	3275
5		Illite/Muscovite	10	1	217
6		Mg-Carbonate	10	1	4856
7		Plagioclase	10	1	1046
8		Prehnite	10	1	1150
9		Serpentine	10	1	6155
Total			90	9	26611

channels. The false-color image and ground-truth image of the NF dataset are shown in Fig. 6, which mainly contains nine mineral categories: fe-olivine, epidote, chlorite, bassanite, illite/muscovite, mg-carbonate, plagioclase, prehnite, and serpentine. The category name, detailed sample number, and labeled color are given in Table II.

3) Utopia Planitia: UP is the landing site of the Tianwen-1 Mars rover Zhurong, located in the southern part of UP. This area is one of the largest impact basins in the solar system and has important scientific exploration value. The spatial resolution of this dataset is 478 × 595, containing 432 bands. The false-color image and ground-truth image of the UP dataset are shown in Fig. 7. There are nine main mineral classifications. The class name, detailed sample number, and labeled color are given in Table III.

Furthermore, for each category of each dataset, we create a mask to select pixels belonging to that category, extract the spectral data of these pixels, calculate the average spectrum, and plot the average spectrum curve for each dataset. Fig. 8



(a) (b)

Fig. 7. UP dataset. (a) False-color image. (b) Ground-truth map.

TABLE III
CATEGORY NAMES AND SAMPLE NUMBERS IN THE UP DATASET

Class ID	Color	Name	Train	Val	Test
1		Analcime	10	1	1740
2		Bassanite	10	1	5796
3		High-Ca Pyroxene	10	1	421
4		Illite/Muscovite	10	1	261
5		Low-Ca Pyroxene	10	1	6763
6		Mg-Smectite	10	1	1155
7		Monohydrated sulfate	10	1	329
8		Plagioclase	10	1	191
9		Prehnite	10	1	583
Total			90	9	17239

shows the analysis results of the average spectral reflectance curves of mineral categories in the three datasets. Through in-depth analysis of spectral features, it can be found that although these different minerals have essential differences in genesis and composition, their spectral curves show obvious similarity characteristics. The significant interclass similarity in this high-dimensional feature space will not only lead to a decrease in the discrimination ability of traditional classification algorithms, but will also greatly increase the complexity requirements of the classification model, thereby significantly increasing the technical difficulty of the hyperspectral mineral classification task. For these three datasets, different colors in each dataset represent different minerals.

We adopt the multi-index weighted fusion method to calculate the band importance by comprehensively considering four indicators: interclass statistical separability, variance difference, information gain, and band correlation. By calculating the band importance of the three datasets, we discovered a significant common pattern: the most distinctive bands of all datasets are concentrated in the long-wave region (2.3–3.9 μm). Specifically, the important band range of the Holden dataset is 374–439, which is consistent with its paleolacustrine environment. Lacustrine sediments typically contain complex combinations of altered minerals and require a wider spectral range for effective distinction. The important band range of the NiliFossae dataset is 398–424, which may reflect the geological feature dominated by carbonate minerals in this area. The important band range of the Utopia dataset is 395–431, which is consistent with the mixed mineral environment of its impact basin. We placed the most distinctive bands of each dataset in Fig. 9.

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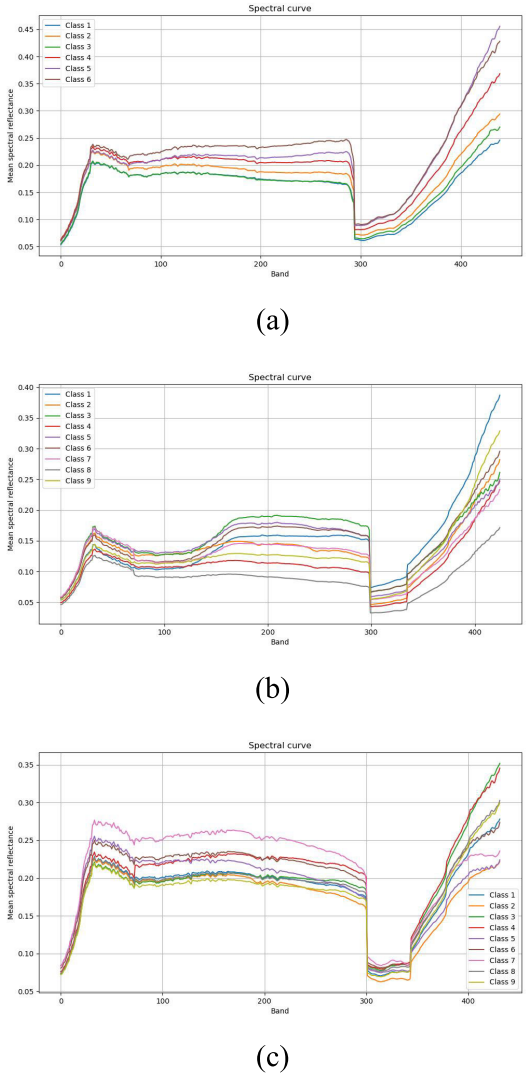


Fig. 8. Mean spectrum curve. (a) HC. (b) NF. (c) UP.

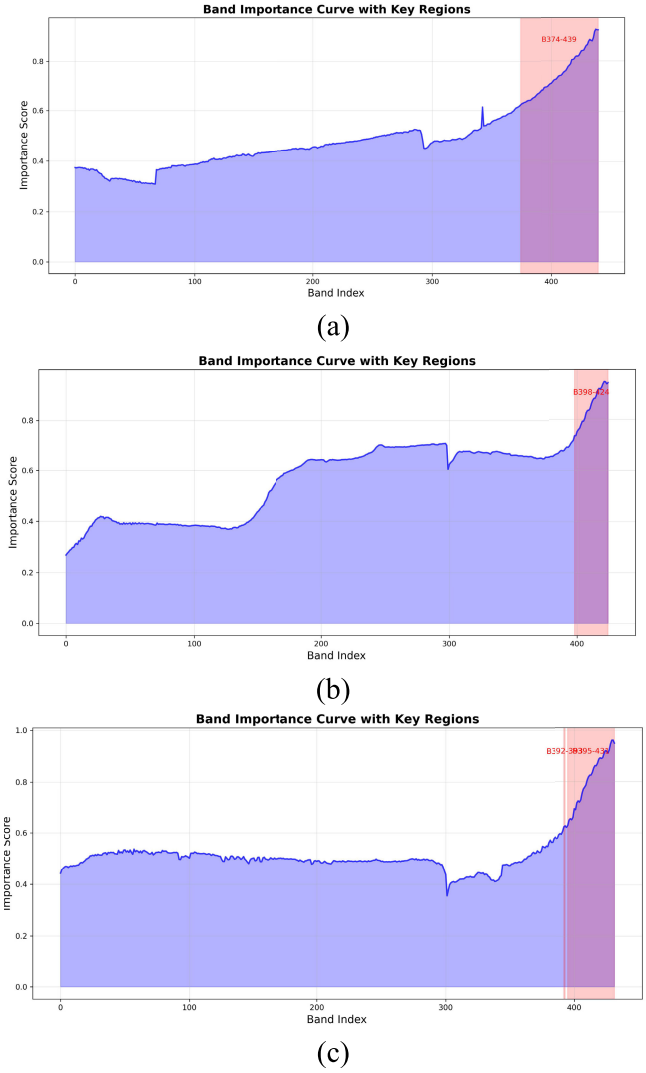


Fig. 9. Most distinguishable bands in the dataset. (a) HC. (b) NF. (c) UP.

TABLE IV
KEY PARAMETERS AND THEIR CONFIGURATIONS

Parameter	Symbol	Range/Value	Description
Spectral kernel width	ω_1	$[100, 1000]$	Eq. (5)
Spatial kernel width	ω_2	$[100, 1000]$	Eq. (6)
ContraNorm temperature	τ	1.0	Eq. (15)
ContraNorm scale	s	0.1	Eq. (10)
Fusion weight	α	$[0, 1]$, init. 0.5	Eq. (27)
Learning rate		5×10^{-4}	Adam optimizer

During the experiment, we used the following key parameters. Table IV lists the parameters we employed. This article selects three evaluation indicators: overall accuracy (OA), average accuracy (AA), and kappa statistic (kappa) to evaluate the experimental results.

B. Comparative Experiment

To verify the effectiveness of the proposed HDCGNet, we conducted a comparative analysis with the established CNN-based spectral–spatial classifiers. The more classic HybridSN [48] and A^{2}S^{2}K-ResNet [25] that use the attention mechanism of CNN are used for comparison, and AMGCFN [46] based on the attention mechanism of GCN is also selected. Second, we also analyzed the performance of transformer-based models such as SSFTT [49] and SPFormer [50], which focus on global information extraction. In addition, the dual-branch model CEGCN [30] and MCTGCL [7] are also compared. Finally, we also selected the currently popular MambaHSI [51] based on the Mamba model and SSEFN [52] based on the small sample model for comparative experiments. For the training process, ten samples were randomly selected from each category. To ensure the reliability of our experimental results, ten independent runs were performed for each method and dataset, and the AA and standard deviation are reported in Tables V–VII. The highest accuracy of each row is marked in bold to facilitate easier comparison and interpretation of the results.

To ensure fairness, all experiments were conducted using the publicly available codes of the relevant authors. Tables V–VII record the classification results of our HDCGNet model and other comparison models on the HC, NF, and UP datasets.

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TABLE V
CLASSIFICATION PERFORMANCE OF VARIOUS METHODS ON THE HC DATASET (%)

NO.	HybridSN	A^2S^2K-ResNet	CEGCN	SSFTT	SPFormer	MambaHSI	SSEFN	AMGCFN	MCTGCL	OURS
1	61.34±24.13	85.32±16.43	96.97±3.54	**99.96±0.07**	99.01±1.57	95.72±3.22	95.72±4.80	77.08±38.87	98.35±2.37	95.32±7.12
2	67.72±18.09	82.09±10.11	81.30±10.33	82.03±6.41	89.06±6.89	85.74±7.75	**91.63±5.07**	77.04±21.96	87.66±8.35	89.14±6.60
3	55.24±14.07	70.65±11.56	83.07±11.29	**91.10±2.51**	89.97±5.19	80.97±4.87	81.80±8.16	71.96±21.01	87.12±8.65	88.81±8.18
4	95.62±1.60	**97.48±4.63**	82.86±8.28	71.54±1.39	90.08±8.89	96.54±2.66	85.84±5.59	93.42±4.80	87.76±5.69	93.26±4.93
5	98.15±4.61	**98.34±2.45**	92.42±4.23	82.39±8.37	89.50±10.47	90.25±7.27	90.46±6.83	95.55±2.96	95.98±2.22	95.46±2.20
6	54.92±22.23	68.82±27.46	**99.41±0.63**	97.61±0.17	96.96±4.15	99.31±0.63	99.21±0.68	96.48±7.42	99.04±0.85	99.39±0.58
OA	84.04±10.13	90.97±5.17	87.93±4.01	80.43±3.71	90.32±5.99	91.84±3.81	88.71±2.46	90.80±5.57	92.08±2.37	**93.81±1.91**
AA	72.16±12.02	83.79±7.44	89.34±3.78	87.44±1.88	92.43±2.73	83.33±1.86	87.30±1.30	85.25±13.30	92.65±1.49	**93.57±1.63**
Kappa	77.91±12.83	87.07±6.89	82.66±5.65	72.61±4.78	86.40±7.95	81.97±3.01	83.44±3.63	86.48±8.40	88.43±2.57	**90.97±2.74**

TABLE VI
CLASSIFICATION PERFORMANCE OF VARIOUS METHODS ON THE NF DATASET (%)

NO.	HybridSN	A^2S^2K-ResNet	CEGCN	SSFTT	SPFormer	MambaHSI	SSEFN	AMGCFN	MCTGCL	OURS
1	**96.87±4.07**	96.72±2.43	88.51±5.68	49.62±3.81	93.02±6.26	90.78±4.56	8.24±4.21	90.06±7.33	86.18±7.96	91.18±6.07
2	62.52±11.18	69.69±12.17	93.24±4.49	**97.30±1.41**	86.71±6.78	84.44±7.51	93.09±5.16	83.24±18.47	87.07±5.72	90.19±6.60
3	49.37±12.82	63.47±15.59	92.01±4.00	91.84±2.13	**92.93±3.90**	86.40±5.73	92.42±4.65	86.14±17.32	91.62±4.17	91.59±4.61
4	72.43±6.94	86.04±5.55	87.46±7.55	69.81±2.26	85.72±6.92	82.17±8.36	80.78±10.41	88.23±11.87	76.55±6.02	**92.04±7.34**
5	24.93±10.76	35.91±8.24	80.87±13.77	82.40±6.37	79.04±13.68	81.61±8.47	82.21±11.01	83.98±14.85	80.09±12.38	**86.55±13.33**
6	**89.78±10.80**	89.70±8.26	89.30±6.56	66.57±3.20	85.86±5.15	79.85±9.42	83.98±7.60	88.60±9.18	66.94±8.94	89.65±6.92
7	72.15±12.31	78.78±13.57	84.96±12.01	89.16±2.53	87.86±5.41	84.64±7.68	**92.77±5.04**	80.76±8.22	84.13±7.07	89.30±10.59
8	53.09±15.34	66.56±12.50	88.18±4.34	83.44±2.56	**89.21±3.60**	72.37±14.14	83.93±7.79	83.93±10.36	84.38±7.47	87.71±7.13
9	**92.87±6.02**	90.43±6.89	79.91±9.01	74.21±6.86	88.85±4.63	87.70±4.04	88.28±7.23	82.30±7.72	85.36±10.67	87.47±9.65
OA	81.46±4.91	86.29±3.41	86.54±3.18	66.53±1.40	89.14±2.60	85.60±2.44	86.73±3.00	86.79±5.10	81.20±3.65	**89.85±2.94**
AA	68.22±4.76	75.26±3.70	87.16±2.84	78.26±1.42	87.69±1.63	83.33±1.86	87.30±1.30	85.25±5.16	82.68±3.09	**89.52±2.33**
Kappa	77.01±6.01	82.87±4.14	83.16±3.97	59.51±1.58	86.43±3.17	81.97±3.01	83.44±3.63	83.43±6.36	76.57±4.45	**87.21±3.65**

TABLE VII
CLASSIFICATION PERFORMANCE OF VARIOUS METHODS ON THE UP DATASET (%)

NO.	HybridSN	A^2S^2K-ResNet	CEGCN	SSFTT	SPFormer	MambaHSI	SSEFN	AMGCFN	MCTGCL	OURS
1	**91.34±3.77**	89.40±10.73	81.73±4.71	66.23±2.25	77.94±6.19	84.02±7.38	74.50±9.02	85.50±11.91	77.20±6.49	90.48±6.33
2	83.35±6.40	88.63±6.31	86.17±7.35	80.02±4.18	**93.24±3.35**	92.88±2.99	90.16±5.21	84.95±7.52	83.93±6.75	93.00±3.33
3	85.29±4.80	85.99±6.35	92.53±10.16	91.05±1.41	92.63±9.15	**94.32±6.39**	82.89±6.92	80.21±28.32	92.20±6.77	86.51±14.57
4	71.15±9.58	63.13±18.97	99.00±0.99	91.92±4.39	98.32±3.38	96.36±4.41	96.13±5.07	88.01±29.42	94.77±6.26	**99.16±0.93**
5	**97.78±1.44**	96.63±2.23	86.95±4.51	97.19±2.62	90.75±7.99	89.03±7.30	88.30±6.77	89.60±7.13	89.04±6.86	88.15±6.27
6	80.14±10.47	88.09±11.76	86.10±8.82	**97.47±1.15**	89.39±11.22	90.78±5.52	94.70±4.77	91.21±9.40	93.43±7.16	88.07±10.46
7	91.66±9.76	92.37±18.75	**100.00±0.00**	99.79±0.64	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	99.91±0.27	**100.00±0.00**
8	44.92±13.34	50.89±12.39	93.38±7.69	84.08±5.06	94.06±9.75	92.20±8.38	**94.18±5.88**	83.22±14.65	83.49±17.48	88.28±14.73
9	77.69±9.01	**91.97±7.88**	90.57±4.33	76.96±1.34	85.36±10.11	83.31±8.19	82.79±6.81	85.58±28.94	83.36±4.67	91.76±4.96
OA	87.04±2.76	89.35±2.28	86.87±2.79	87.30±1.68	90.39±3.12	90.23±2.77	88.05±3.19	87.47±4.14	86.54±3.09	**90.49±2.16**
AA	80.37±2.88	83.01±3.25	90.71±1.53	87.19±1.09	91.30±1.49	91.43±1.12	89.29±1.90	87.59±12.56	88.59±1.62	**91.71±2.26**
Kappa	82.08±3.78	85.31±3.11	82.01±3.67	82.50±2.21	86.69±4.27	86.50±3.76	83.56±4.24	82.79±5.72	81.69±3.89	**86.81±2.94**

Figs. 10–12 show the classification graphs of different models on the three datasets. From the experimental results on the three benchmark Mars HSI datasets, we can draw the following observations.

1) Comparative Analysis: For fair comparison, all HSI classification methods are run on the same training set and test set. The divided training set, validation set, and test set are shown in Tables I–III. Since the HC dataset has a higher spatial resolution, more retained bands, and relatively fewer categories, its classification accuracy is relatively higher than that of other datasets. According to the results reported in Table IV, the accuracy of HDCGNet is significantly higher than that of other methods. Among them, HDCGNet has the highest OA, reaching 93.81%, which is 9.77% (HybridSN), 2.84% (A^2S^2K-ResNet), 5.88% (CEGCN), 13.38% (SSFTT), 3.49% (SPFormer), 1.97% (MambaHSI), 5.10% (SSEFN), 3.01% (AMGCFN), and 1.73% (MCTGCL) higher than other methods. In addition, AA and Kappa of HDCGNet are also significantly improved.

Although the NiliFossae dataset has a high spatial resolution, the irregular distribution of objects and the overlapping categories make classification difficult, and the classification accuracy is at a lower level compared with other datasets. According to the results reported in Table V, the accuracy of HDCGNet is significantly higher than that of other methods. Among them, HDCGNet has the highest OA, reaching 89.85%, which is 8.39% (HybridSN), 3.56% (A^2S^2K-ResNet), 3.31% (CEGCN), 23.32% (SSFTT), 0.71%

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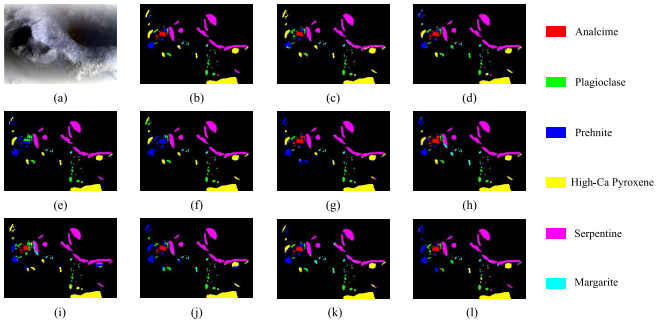


Fig. 10. Classification diagrams obtained by various algorithms on the HC dataset. (a) HC dataset. (b) Ground-truth. (c) HDCGNet. (d) MCTGCL. (e) AMGCFN. (f) SSEFN. (g) MambaHSI. (h) SPFormer. (i) SSFTT. (j) CEGCN. (k) A^{2}S^{2}K-ResNet. (l) HybridSN.

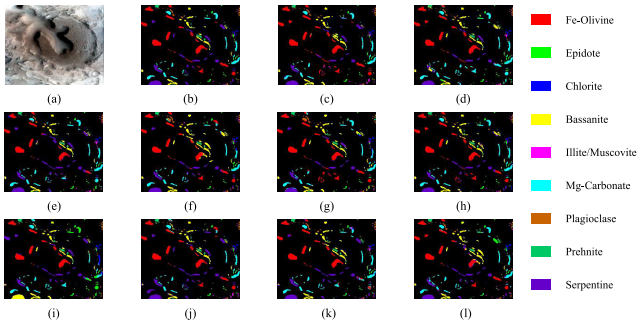


Fig. 11. Classification diagrams obtained by various algorithms on the NF dataset. (a) NF dataset. (b) Ground-truth. (c) HDCGNet. (d) MCTGCL. (e) AMGCFN. (f) SSEFN. (g) MambaHSI. (h) SPFormer. (i) SSFTT. (j) CEGCN. (k) A^{2}S^{2}K-ResNet. (l) HybridSN.

(SPFormer), 4.25% (MambaHSI), 3.12% (SSEFN), 3.06% (AMGCFN), and 8.65% (MCTGCL) higher than other methods. In addition, AA and kappa of HDCGNet are also significantly improved.

The UP dataset has a high spatial resolution, and the smooth and regular distribution of objects makes classification simple and has a high classification accuracy. According to the results reported in Table VI, the accuracy of HDCGNet is significantly higher than that of other methods. Among them, HDCGNet has the highest OA, reaching 90.49%, which is 3.45% (HybridSN), 1.14% (A^{2}S^{2}K-ResNet), 3.62% (CEGCN), 3.19% (SSFTT), 0.10% (SPFormer), 0.26% (MambaHSI),2.44% (SSEFN), 3.02% (AMGCFN), and 3.95% (MCTGCL) higher than other methods. In addition, the AA and kappa of HDCGNet are also significantly improved.

2) Comparative Analysis: The visual comparison of the classification results is shown in Figs. 10–12. The existing classification models generally have obvious misclassification phenomena. There are a large number of discrete misclassified pixels within each category area. Second, the boundaries of the transition area between categories are unclear. In addition, some ground object categories are seriously confused. In contrast, the HDCGNet method proposed in this article shows significant advantages on all test datasets: the number of misclassified pixels in the classification result map is the least; the consistency within each category area is the highest; the boundaries between classes are the clearest. These visual qualitative analysis results are highly consistent with the quantitative evaluation indicators in Tables V–VII. The classification map is obviously less noisy and smoother, which proves the superiority of our method.

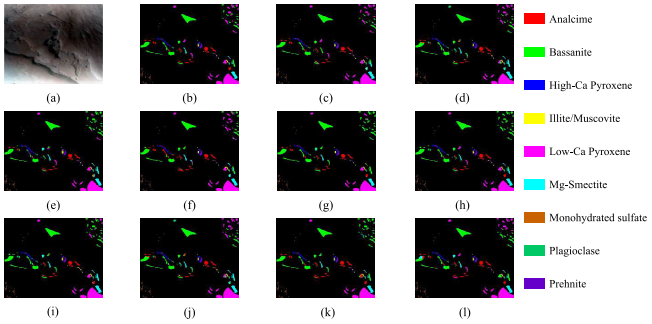


Fig. 12. Classification diagrams obtained by various algorithms on the UP dataset. (a) UP dataset. (b) Ground-truth. (c) HDCGNet. (d) MCTGCL. (e) AMGCFN. (f) SSEFN. (g) MambaHSI. (h) SPFormer. (i) SSFTT. (j) CEGCN. (k) A^{2}S^{2}K-ResNet. (l) HybridSN.

3) Comparative Model Analysis: The CNN-based HSI classification models (i.e., HybridSN and A^{2}S^{2}K-ResNet) show excellent performance on the three datasets because they fully utilize the local spatial–spectral features of HSI. The GCN-based HSI classification model (i.e., AMGCFN) achieves excellent performance on different HSI data, captures nonlocal spatial dependencies, and effectively handles irregular ground object distribution. The transformer-based HSI classification models (i.e., SSFTT and SPFormer) focus on global information extraction. The dual-branch HSI classification models (i.e., CEGCN and MCTGCL) are mainly due to their effective exploration of the local and global spatial–spectral features of HSI by combining CNN and GCN or transformer dual-branch parallel processing. The HSI classification model based on Mamba (i.e., MambaHSI) mainly integrates spatial and spectral information by leveraging its excellent long-range modeling capabilities. The HSI classification model for small samples (i.e., SSEFN) enhances feature discrimination ability and learns with fewer samples. The HDCGNet method proposed in this article aggregates the spatial and spectral features of HSI to further improve the classification performance.

C. Experimental Settings

To maximize the effectiveness of the proposed network, we implemented a detailed experimental configuration. We set the total number of training sets to 500, the number of hyperpixel nodes to 500, and the learning rate to 0.0005, following standard practice. All experimental results are calculated on the basis of the average of the results after ten independent runs of randomly selected initial training samples.

1) Sample Size: In order to evaluate the effectiveness of the proposed method under different labeled samples, we randomly selected 5, 10, 15, 20, and 25 annotated samples and conducted ten experiments. Under the same circumstances, we found that ten annotated samples had the best OA value, so this article uniformly chose to use ten annotated samples for

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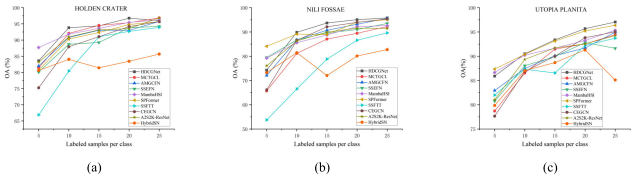


Fig. 13. OA (%) of different methods on three datasets with different numbers of labeled samples. (a) HC. (b) NF. (c) UP.

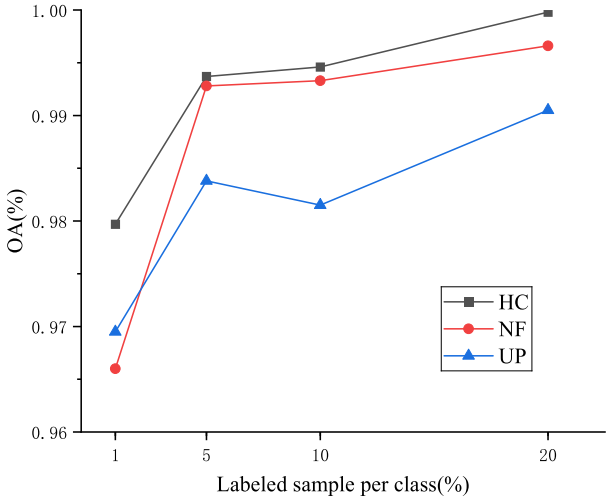


Fig. 14. OA rate (%) of different methods on three datasets with a larger number of labeled samples.

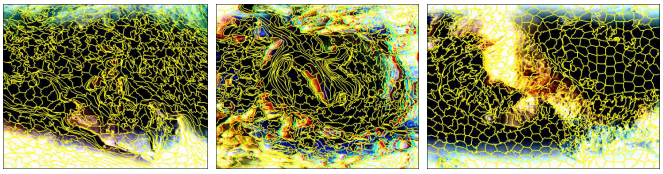


Fig. 15. Hyperpixel image with $\lambda = 500$. (a) HC. (b) NF. (c) UP.

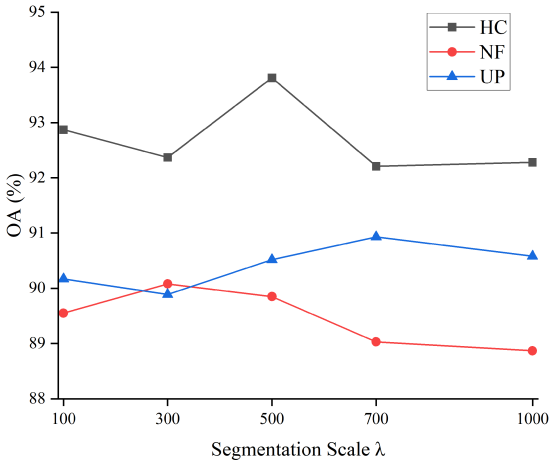


Fig. 16. OA under different λ values on each dataset.

experiments. The average OA in the three datasets is shown in Fig. 13. In addition, we have also added experiments with larger sample sizes with annotations, setting 1%, 5%, 10%, and 20% of the total sample size, respectively. This also proves that the advantages of HDCGNet do not merely stem from an extremely low data volume environment, but rather from a scalable and robust model. The average OA on the three sets of datasets is shown in Fig. 14.

2) λ: In order to determine the optimal segmentation scale and evaluate the impact of the partition size of SLIC, we keep other parameters unchanged and make λ systematically vary in the range of [100, 300, 500, 700, 1000]. Fig. 15 shows the hyperpixel images of the three datasets when $\lambda = 500$. Specifically, a larger scale will produce a smaller graph, resulting in fewer hyperpixel nodes and larger hyperpixels, which can suppress noise. On the contrary, a smaller scale will produce a larger graph composed of small hyperpixels with noise. Fig. 16 shows the relationship between the segmentation scale and OA. When the segmentation scale λ of the three datasets is set to 500, 300, and 700, respectively, the OA value is the highest. In order to maintain relative fairness in the experimental results of this article, λ is uniformly set to 500.

3) K: In order to determine the optimal number of neighbor nodes, K varies in the range of [3, 6, 9, 12, 15]. As shown in Fig. 17, the relationship between OA and the number of neighbor nodes. When the K value is equal to 12, the OA of the HC dataset reaches the highest; for the NF dataset, in the range of [6, 9, 12, 15], OA has an upward trend. For the UP dataset, when the number of neighbor nodes is in the range of [3, 6, 9, 12, 15], OA is relatively insensitive, and the node is selected in the 2-hop neighbor, and the coverage space range is not large, so its decline will not be much lower than the optimal OA. After comprehensive consideration, the K values of the three datasets are set to 12, 15, and 9, respectively.

4) Epoch: We use five different training epochs ranging from 100 to 500 to evaluate the performance of our proposed HDCGNet method. Fig. 18 shows the classification performance of our HDCGNet method with different training times on the three datasets. We can see that with the increase in the number of iterations, the OA of HDCGNet on the three datasets shows an upward trend. Moreover, HDCGNet converges with the increase in training epochs, and finally reaches 93.81%, 89.85%, and 90.52% OA values at 500 epochs on the three datasets, respectively. Since the training time tends to increase with the increase in epochs, we set the training epoch of HDCGNet to 500 to balance the training time and performance. In order to avoid unfair comparison, considering the undertraining of the compared models, we set the training epoch of the compared methods to 500.

5) Learning Rate: In this section, we study the effects of different learning rate settings on the proposed HDCGNet. The learning rate determines the model's convergence and convergence efficiency. Several commonly used learning rate settings are empirically tested and qualitatively compared on three datasets. The experimental results of different learning rate settings on the three datasets are shown in Fig. 19. We can see that when the learning rate is 0.05, the OA of the three datasets is extremely low, and as the learning rate decreases,

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TABLE VIII
ABLATION CLASSIFICATION PERFORMANCE OF THE PROPOSED METHOD ON HC, NF, AND UP DATASETS (%)

Dataset	Case	DSC	CN	ACAFM	OA	AA	Kappa
HC	1				90.35±5.80	83.89±17.27	85.80±8.80
	2	✓			90.06±4.52	87.48±7.04	85.68±6.28
	3	✓	✓		91.82±3.18	88.94±7.15	88.12±4.56
	4	✓	✓	✓	**93.81±1.91**	**93.57±1.63**	**90.97±2.74**
NF	1				87.46±7.81	85.78±8.09	84.40±9.42
	2	✓			89.02±3.04	87.87±3.45	86.23±3.77
	3	✓	✓		87.27±5.47	84.13±10.79	83.89±6.94
	4	✓	✓	✓	**89.85±2.94**	**89.52±2.33**	**87.21±3.65**
UP	1				88.89±3.20	88.91±6.84	84.80±4.29
	2	✓			90.38±2.46	91.57±3.05	86.74±3.33
	3	✓	✓		89.26±2.19	90.65±6.25	85.19±2.95
	4	✓	✓	✓	**90.49±2.16**	**91.71±2.26**	**86.81±2.94**

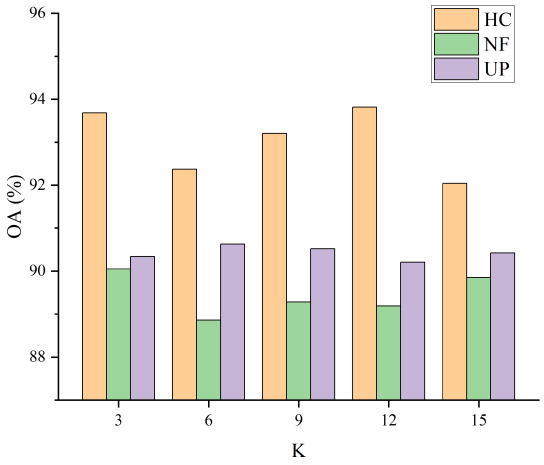


Fig. 17. OA under different K values on each dataset.

the OA tends to increase continuously. Therefore, we set the learning rate to 0.0005 to report all experimental results.

D. Ablation Experiment

As described in Section III, the proposed method mainly consists of the DSC module, the ContraNorm normalization layer (hereinafter referred to as CN), and CAFM feature fusion. To evaluate the effectiveness of each component, we conducted ablation studies on these specific modules, and the hardware used for training was the NVIDIA GeForce RTX 4090 GPU.

First, we define a baseline network that excludes DSC, CN, and ACAFM. The baseline consists of a CNN-based branch for local feature extraction and a GCN-based branch for parallel processing of global features. Then, we gradually add DSC and CN by replacing the existing modules in the baseline with these components. Finally, we fuse ACAFM to form the final complete architecture. The four experimental cases are as follows.

Case 1: Network without DSC, CN, and ACAFM.
Case 2: Network without CN and ACAFM.
Case 3: Network without ACAFM.
Case 4: Network including DSC, CN, and ACAFM.

For each class, we randomly select ten labeled samples and conduct ten experiments independently. Table VIII summarizes the average classification performance of the three datasets as well as the standard deviation. Undoubtedly, the best results of OA, AA, and Kappa are highlighted in bold. It is obvious from Table VIII that our proposed DSC, CN, and ACAFM can significantly improve OA, AA, and Kappa. Specifically, DSC effectively captures the local features of HSI with its lower complexity. ContraNorm helps generate more evenly distributed features by directly applying losses to the graph node representation. Meanwhile, ACAFM extracts spectral–spatial features from the pixel level and hyperpixel level, respectively, through relatively parallel multiscale convolutional subnetworks and graph node subnetworks, and performs feature fusion, thereby comprehensively capturing the multilevel contextual semantic features of HSIs.

In addition, in the HC dataset, we conducted ablation experiments for three hyperedge composition strategies. Each of these strategies was deleted separately to showcase the contribution of each strategy. The ablation results are shown in Table IX.

E. Comparative Analysis of kNN Hyperedge Composed of SSIM

To demonstrate that SSIM can comprehensively reflect the statistical dependence and structural consistency of spectral shapes, we use SSIM similarity and compare it with Euclidean distance, cosine similarity, and Manhattan distance to construct the hyperedge of kNN.

SSIM maintains the structural correlation of features within the neighborhood, which helps to maintain spatial continuity and local consistency in the construction of hypergraphs. The Euclidean distance only measures the amplitude difference between two eigenvectors and is highly sensitive to scale and illumination changes. In Mars hyperspectral data, it is easily affected by noise and reflection intensity. Cosine similarity measures the directional consistency but ignores the local intensity and contrast information. It measures the sum of the absolute differences of spectral features in each dimension, emphasizing the consistency of the overall contour of the

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TABLE IX
ABLATION STUDY OF GRAPH CONSTRUCTION STRATEGIES ON HC DATASET (%)

Dataset	Case	$\varepsilon_{\text{pair}}$	$\varepsilon_{\text{k-hop}}$	ε_{kNN}	OA	AA	Kappa
HC	1	✓			93.68 ± 2.48	91.45 ± 6.68	90.76 ± 3.65
	2		✓		91.65 ± 3.39	88.00 ± 9.91	87.78 ± 4.95
	3			✓	91.77 ± 3.53	86.94 ± 10.62	87.96 ± 5.17
	4	✓	✓		91.91 ± 3.16	88.28 ± 9.68	88.16 ± 4.63
	5	✓		✓	92.11 ± 3.70	88.12 ± 11.28	88.45 ± 5.46
	6		✓	✓	92.25 ± 3.49	88.37 ± 10.47	88.65 ± 5.15
	7	✓	✓	✓	$\mathbf{93.81 \pm 1.91}$	$\mathbf{93.57 \pm 1.63}$	$\mathbf{90.97 \pm 2.74}$

TABLE X
COMPARATIVE EXPERIMENTS USING KNN TO CONSTRUCT HYPEREDGES (%)

Dataset	Type	OA	AA	Kappa
HC	Euclidean distance	91.86 ± 3.12	87.56 ± 10.32	88.07 ± 4.62
	Cosine similarity	91.84 ± 3.17	87.86 ± 9.53	88.08 ± 4.63
	Manhattan distance	92.81 ± 3.85	89.84 ± 8.29	89.50 ± 4.71
	SSIM	$\mathbf{93.81 \pm 1.91}$	$\mathbf{93.57 \pm 1.65}$	$\mathbf{90.97 \pm 2.74}$

spectral curve, but is not sensitive to the nonlinear structure and local contrast changes between dimensions. SSIM decomposes similarity into three parts: luminance, contrast, and structure, which is equivalent to the perceived similarity measured in the structural manifold space, and pays more attention to the relative changes of spectral shapes rather than absolute differences. Therefore, SSIM can comprehensively reflect the statistical dependence and structural consistency of spectral shapes.

Ultimately, through experiments, we found that the OA of using SSIM has improved, verifying its stronger discriminative power and robustness. The results are shown in Table X.

F. Complexity Analysis

In order to systematically evaluate the operating efficiency and computational complexity of the proposed method, we compared and analyzed multiple algorithms from two dimensions: training time and network parameter quantity (Params). The hardware used for training was a NVIDIA GeForce RTX 4090 GPU. Each experiment was repeated ten times to eliminate random errors, and the average training time was finally reported. As shown in Table VIII, compared with other network architectures, the HDCGNet method proposed by us maintains model performance while keeping its training time and computational resource consumption within a reasonable range, showing good computational efficiency and achieving a good balance between model complexity and computational efficiency. The training time and training parameter quantities in the experiment can be seen in Table XI.

Although HDCGNet adopts multistrategy hypergraphs (pairwise, k-hop, and kNN), the additional computational overhead has been strictly controlled. Since it is first segmented into hyperpixels, the number of subsequent nodes is significantly reduced. Using the SLIC algorithm, the complexity is

$$O\left(N \cdot \log N\right)$$

where N is the number of pixels.

During the process of generating hyperedges, the computational overhead is as follows.

1) *K-Hop Hyperedge:* Generated by BFS/DFS based on adjacency matrix, with a complexity of $O(N \cdot k)$.
2) *kNN Hyperedge:* Based on the precalculated SSIM distance matrix ($O(N^2 \cdot d)$), perform the top-K search ($O(N \cdot \log k)$).
3) *Paired Hyperedges:* Directly transformed from the basic adjacency matrix, with a complexity of $O(E)$.

The total complexity is approximately $O(N^2 \cdot d + N \cdot \log k + N \cdot k)$, where N is the number of hyperpixels (much less than the number of pixels $H \times W$). Since the image is segmented into hyperpixels rather than individual pixels, the total cost is still very low compared with CNN feature extraction, and the overall overhead is controllable and can be ignored.

The complexity of hypergraph convolution during the training process can be regarded as the HGNNConv layer. The computational complexity of each layer is approximately

$$O\left(E + N \cdot F\right)$$

similar to the standard GCN convolution, where E is the number of hyperedges and F is the feature dimension. The overall complexity is comparable to that of GCN, without introducing excessive additional overhead.

G. Visualization Analysis of the Training Process

As shown in Fig. 20, in order to evaluate the convergence and stability of the model, we plotted the loss function curve during the training process. The loss function curve indicates that as the number of training rounds increases, both the training loss and the validation loss show a downward trend and eventually stabilize, suggesting that the model has the best generalization ability at this point. The gap between the two curves is small, indicating that the model does not show significant overfitting. The shaded area represents the standard deviation of multiple random seeds, and the narrower the shade, the better the stability of the model.

Second, to evaluate the quality of the feature representation learned by the model, we used the t-SNE dimensionality reduction technique to visualize the high-dimensional feature space onto a 2-D plane. To enhance the visual clarity and reduce the computational load, we display all the valid pixels. As shown in the visualized graph in Fig. 21, each point represents a test set pixel, and its color corresponds to the real class label. The spatial distance between points reflects

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TABLE XI
RUNNING TIME AND PARAMETER SIZE OF DIFFERENT METHODS ON HC, NF, AND UP DATASETS

Dataset	Metric	HybridSN	A^2S^2K-ResNet	CEGCN	SSFTT	SPFormer	MambaHSI	SSEFN	AMGCFN	MCTGCL	OURS
HC	Params (K)	835.3	750.3	196.3	148.0	95.5	454.9	24903.8	408.8	237.9	291.6
	Train Time (s)	4.65	101.16	29.95	1.82	24.81	153.14	59.25	68.25	64.53	52.10
NF	Params (K)	8076.5	728.3	194.7	148.0	324.0	453.4	24381.6	407.4	238.3	290.8
	Train Time (s)	5.89	144.46	34.80	3.59	50.82	169.54	89.25	78.92	136.44	63.44
UP	Params (K)	8205.6	737.8	195.6	148.0	324.5	454.3	24632.5	408.8	238.3	291.7
	Train Time (s)	6.28	147.78	35.24	3.31	51.18	169.82	88.72	79.43	92.11	62.91

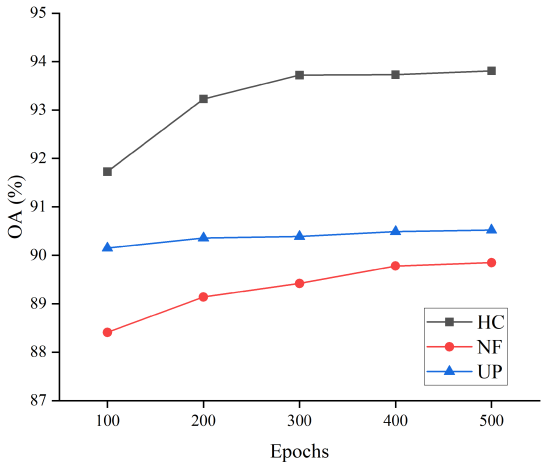


Fig. 18. OA under different epochs on each dataset.

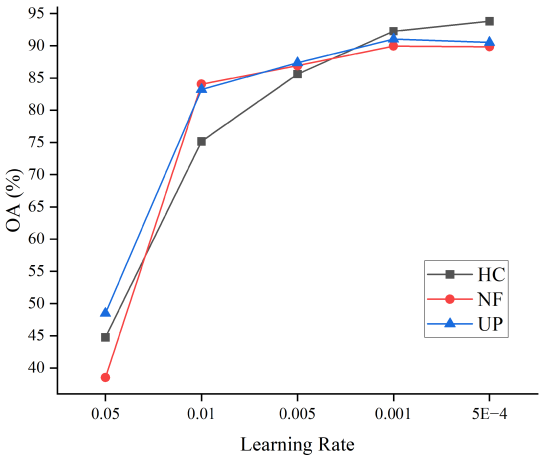


Fig. 19. OA under different learning rates on each dataset.

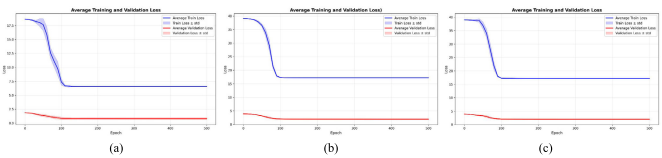


Fig. 20. Average loss curves on each dataset. (a) HC. (b) NF. (c) UP.

the similarity of the category probability distribution output by the model.

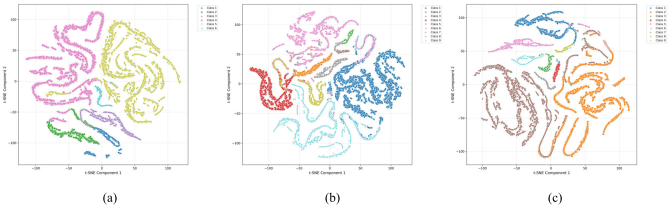


Fig. 21. t-SNE visualization on each dataset. (a) HC. (b) NF. (c) UP.

H. Discussion on the Computational Cost of Hypergraphs

We conducted a detailed analysis of the confusion matrix of the model to determine the mineral categories that are most likely to be misclassified, and discussed the reasons for misclassification from the spectral feature level and the topological structure level, respectively. Among them, the main misclassified categories are as follows.

In the HC dataset, prehnite and serpentine are more likely to be misclassified. In the NF dataset, mg-carbonate and serpentine are more likely to be misclassified. In the UP dataset, bassanite, low-ca pyroxene, and plagioclase are more likely to be misclassified.

The main reasons for our analysis of misclassification are as follows.

1) Reasons at the Spectral Level: The spectra of the misclassified minerals are highly similar. The absorption bands of these misclassified minerals almost completely overlap with those of other minerals in certain bands. As can be seen from Fig. 8, their reflectance curves highly overlap in these bands, resulting in a relatively low spectral resolution. Therefore, the main reason for misclassification is the excessively high spectral similarity, which makes it difficult to distinguish adjacent categories during the feature extraction stage.

2) Topological Reasons: The secondary cause comes from the boundary uncertainty of SLIC segmentation, which leads to blurring of the hyperpixel boundaries, especially in the transitional regions where two minerals coexist. The mixed pixels within these regions may be classified as the same hyperpixel node, thereby introducing local structural noise into the hypergraph structure. However, the multistrategy hyperedge construction proposed in this article can connect adjacent regions through spectral and spatial joint similarity, which can alleviate this problem to a certain extent.

V. CONCLUSION

In this article, we propose a parallel feature fusion network that integrates CNN and GCN for HSI classification, which

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includes two branches (i.e., CNN branch and GCN branch) and a feature fusion module (i.e., ACAFM). First, the CNN branch is employed to extract multiscale features, aiming to capture pixel-level features with differentiated receptive fields through a multiscale feature extraction strategy. The DDR transform is incorporated to achieve efficient DDR of spectral features. Meanwhile, the CMC module is utilized to extract feature representations in the spatial–spectral joint domain by means of multikernel convolution. Second, a special GCN branch architecture is designed, which constructs a hypergraph based on three different hyperedge construction strategies and introduces it into GCN. By integrating the inherent spatial topological structure and spectral features of the segmented region, a joint hypergraph representation model is constructed, and hyperpixel-level deep features with semantic consistency are extracted. Finally, ACAFM uses cross-attention to realize feature interaction between the two branches. The HDCGNet proposed in this article can effectively aggregate spatial–spectral multilevel contextual features at both the pixel and hyperpixel levels, thereby significantly enhancing the classification performance of HSIs. Comparative experiments on three Mars benchmark HSI datasets show that HDCGNet outperforms existing advanced methods in both feature representation learning and cross-level feature fusion.

Although the current method has achieved good classification results in the experimental dataset, we are still committed to further optimizing the performance and efficiency of the model. Future research work will focus on the following directions.

1) Hypergraph Sparsification and Adaptive Pruning: Future research will investigate efficient hypergraph sparsification and adaptive edge selection strategies to reduce redundant connections while preserving topological consistency. This direction can significantly reduce computational complexity and memory cost, enabling real-time processing of large-scale Mars HSI data.

2) Knowledge Distillation for Lightweight Deployment: We plan to explore knowledge distillation from the dual-branch HDCGNet to a single-branch lightweight student model. The goal is to retain the discriminative spectral–spatial representation power of the teacher model while improving inference efficiency on resource-constrained Mars exploration hardware.

3) Multigranularity Alignment and Enhancement: Before feature interaction, design a coarse-to-fine feature alignment module. First, the pixel features are initially aggregated and aligned within the hyperpixel region. Then, through the iterative propagation of the graph neural network, the corrected semantic information of the hyperpixels is fed back to the pixel-level features, forming a bidirectional and iterative refinement process.

4) Adaptive Homophily-Aware Graph Learning [53]: We intend to explore adaptive homophily-aware graph learning. This could be realized through a learnable homophily coefficient that modulates edge weights or even dynamically prunes and adds hyperedges, which would enable the GCN branch to adaptively emphasize semantically meaningful connections while suppressing noisy or spurious links, particularly in regions with mixed pixels or transitional mineral zones.

5) Unsupervised Image Clustering: We plan to explore unsupervised graph learning clustering methods for HSIs to further reduce the model's reliance on scarce labeled samples. By jointly learning node representations and clustering assignments in a graph structure, the model can uncover the inherent similarities between spectral–spatial features in an unsupervised manner. An adaptive graph construction strategy can be introduced to dynamically adjust the node connection relationships and clustering granularity, thereby discovering region structures with semantic consistency.

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